Music Minimal Skip Optimizer

STAT 4830 Slides Week 11

What Changed

Week 11 goal: build a model that incorporates more features and predicts skip for the current event with better context.

- Shifted focus from general next-token modeling to direct skip prediction
  - Token decoding -> classification
- Iterated through 3 transformer model variants to improve target alignment and performance
- Added explainability (permutation importance) and leakage checks for methodological rigor

Backbone Upgrade + Input Representation

- Upgraded transformer capacity: more layers (4 -> 6), higher embedding dimensions (256 -> 768)
- Switched to Pre-RMSNorm
- Added configurable MLP activations (SwiGLU)
- Removed bias

```
song vocab size: 14045
aux numeric features: 30
aux categorical features: 12
```

- Added numeric + categorical auxiliary features.
  - Ex. danceability, acousticness, duration, release year, time of day
- Token = song embedding + numeric projection + categorical embeddings + position.

Performance Progression

Variant A: **first direct-skip emphasis**: reduce song loss weight to 0, emphasis on skip loss.

- skip_f1 = 0.6878

Variant B: **direct per-track skip target**: change input to history up to current song, label becomes skip/no skip using BCE (from joint token for song/skip).

- skip_f1 = 0.7066

Variant C: **direct skip + current event context fusion**: introduce fusion architecture by concatenating three vectors:

[h_hist (transformer history), h_song (embedding of current song), h_evt (embedding of current context)]

- test skip_f1 = 0.8859
- Precision 0.8554, Recall 0.9186 (best epoch 1)

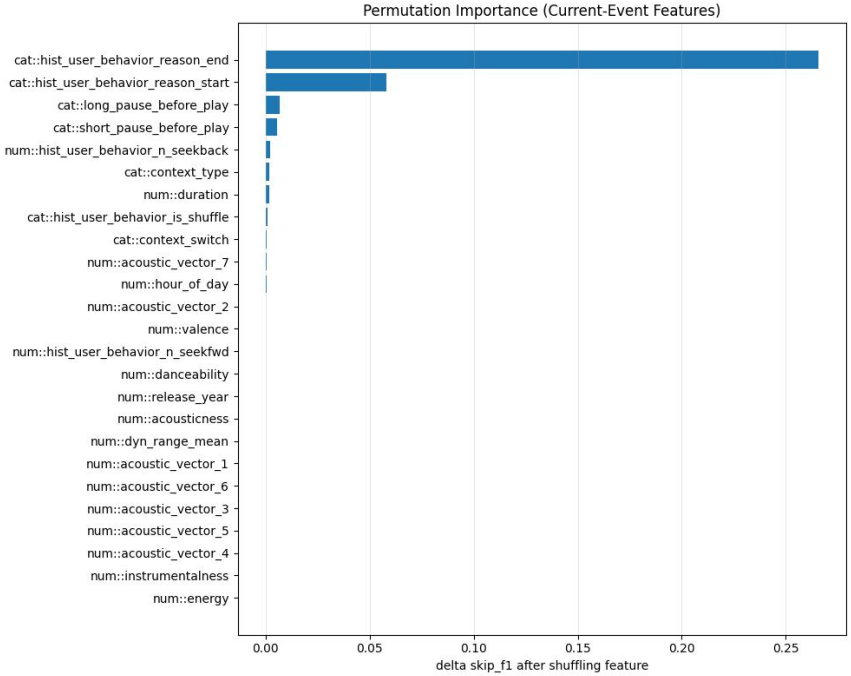
Explainability: Permutation Importance

Permutation importance on context model.

Baseline F1 for this analysis: 0.8763

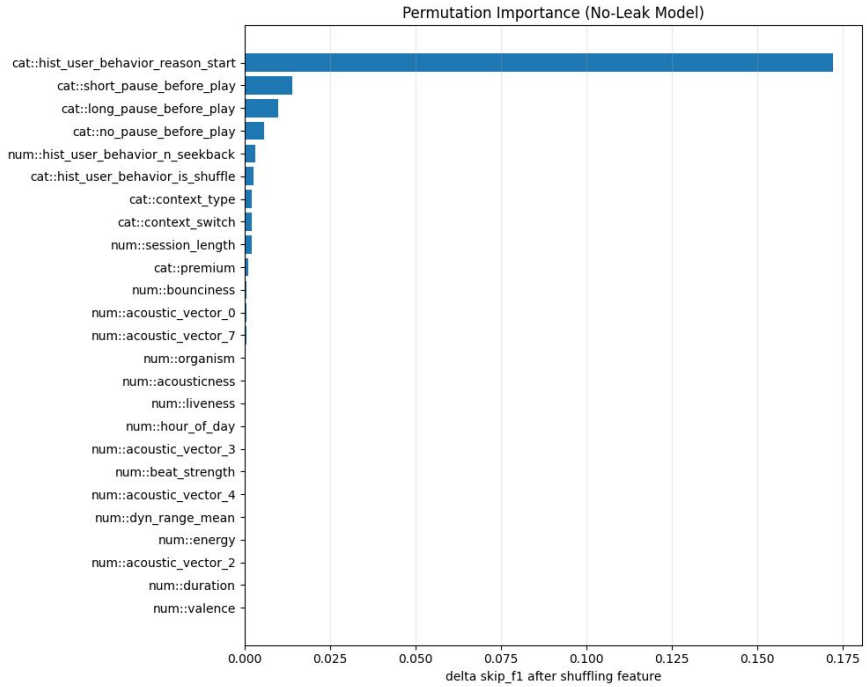
Strongest feature: hist_user_behavior_reason_end

(delta_f1 = 0.2657)



Leakage?

Potential leakage from hist_user_behavior_ reason_end:



Value counts (top 20):

hist_user_behavior_reason_end	count
fwdbtn	81389
trackdone	57701
endplay	14823
backbtn	13129
logout	563
remote	267
clickrow	8

dtype: int64

Cross-tab with skip label (row-normalized):

hist_user_behavior_reason_end	skip=0	skip=1
clickrow	0.0000	1.0000
backbtn	0.1001	0.8999
fwdbtn	0.1900	0.8100
endplay	0.4228	0.5772
remote	0.5468	0.4532
logout	0.7105	0.2895
trackdone	0.9959	0.0041

Final Model

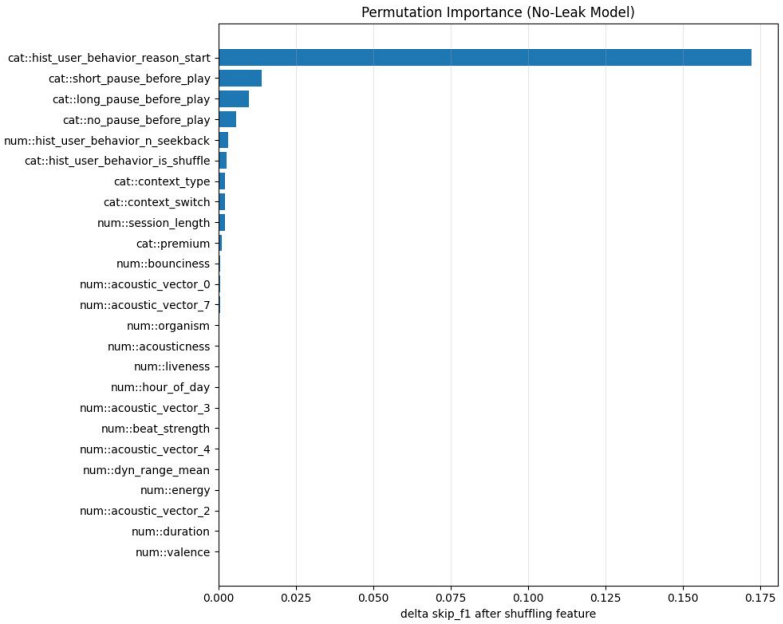
Removed the variable and retrained same architecture:

Best results: test_skip_f1 = 0.8009 (epoch 3, threshold 0.43)

Top no-leak feature: reason_start (delta_f1 = 0.1720)

Cross-tab with skip label (row-normalized):

hist_user_behavior_reason_start	skip=0	skip=1
fwdbtn	0.2352	0.7648
backbtn	0.3145	0.6855
trackerror	0.5109	0.4891
playbtn	0.5161	0.4839
appload	0.5407	0.4593
remote	0.5588	0.4412
endplay	0.6000	0.4000
clickrow	0.6214	0.3786
trackdone	0.8355	0.1645



Next Steps

- Grid search over hyperparameters and pick best val skip f1
- See relative importance of different categories, e.g. audio, context, behavior, time, etc.
- Test the best model over multiple seeds
- Work on sequence generation to minimize skips